

Analysis group meeting

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14 April, 2025



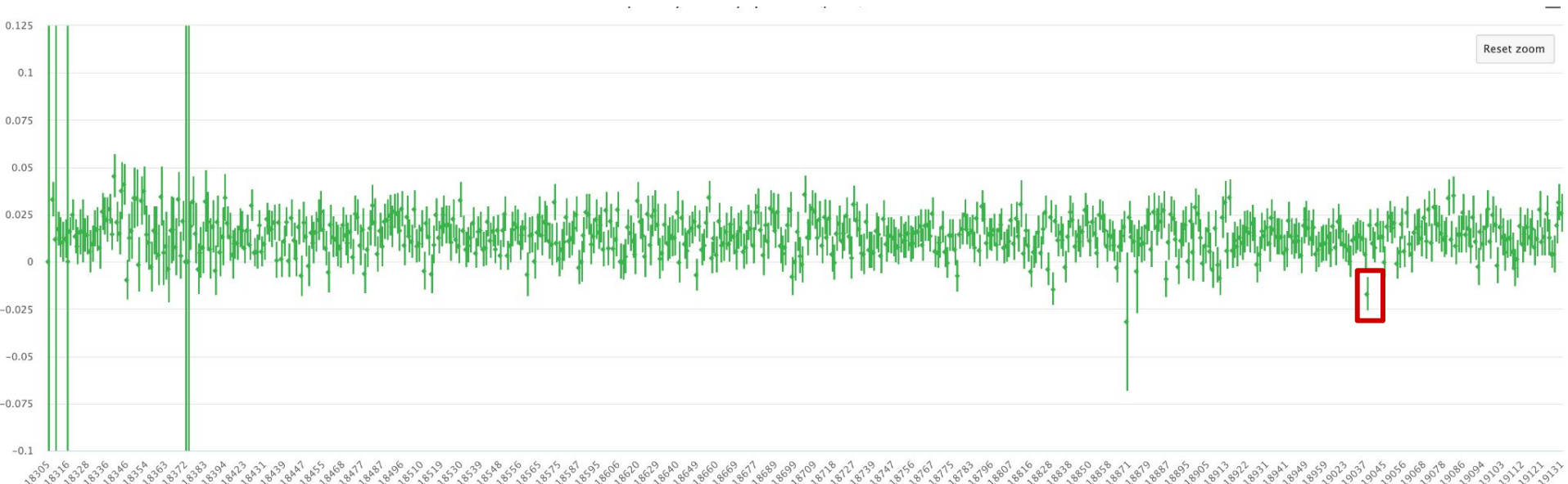
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- Cooking for:
 - training for outbending LD2 network with denoising (train_obLD2_step1_wdenoise/)
 - training for outbending LD2 network without denoising (train_obLD2_step1/)
 - training for inbending LD2 network with denoising (train_ibLD2_step2_wdenoise/)
 - training for inbending LD2 network without denoising (train_ibLD2_step2/)
- Next steps:
 - cook validation data for these four network

High level physics QA timelines

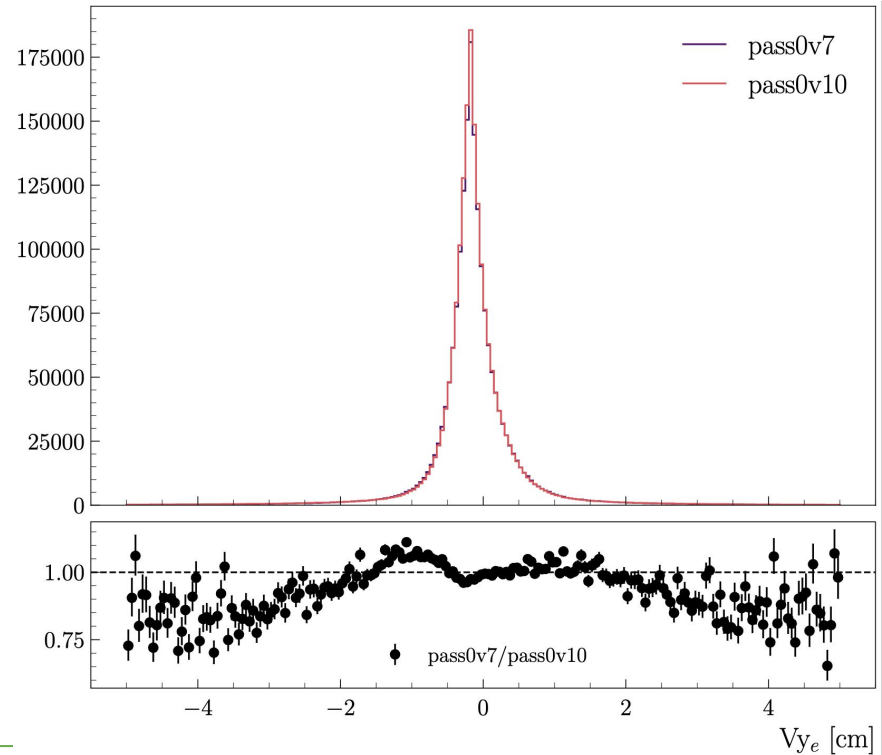
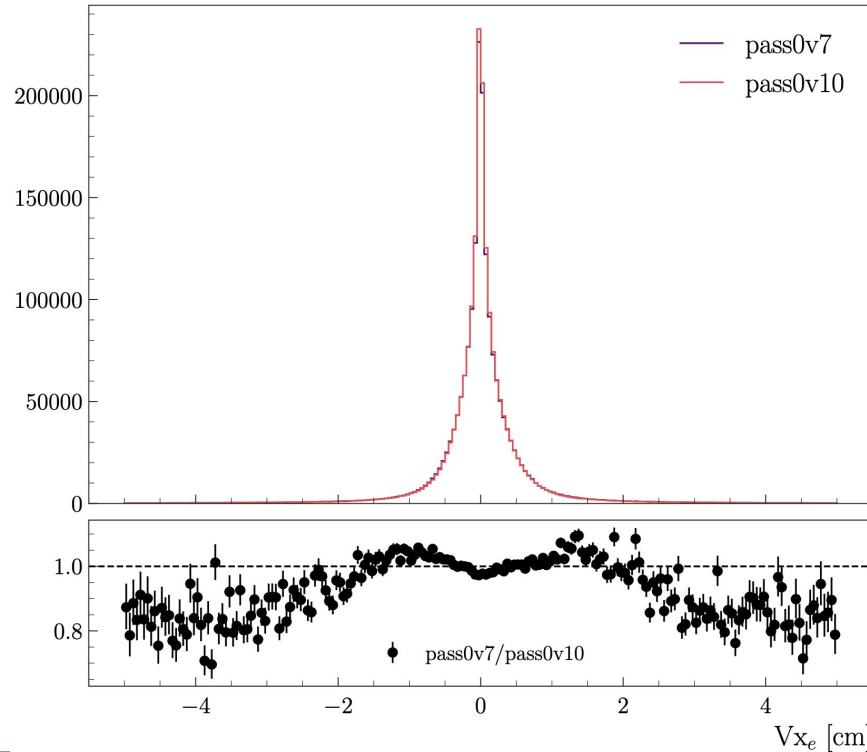
3

- Some strange runs like 19038 which have a negative slope that are close to a change of the HWP: 18886, 18831, 19038

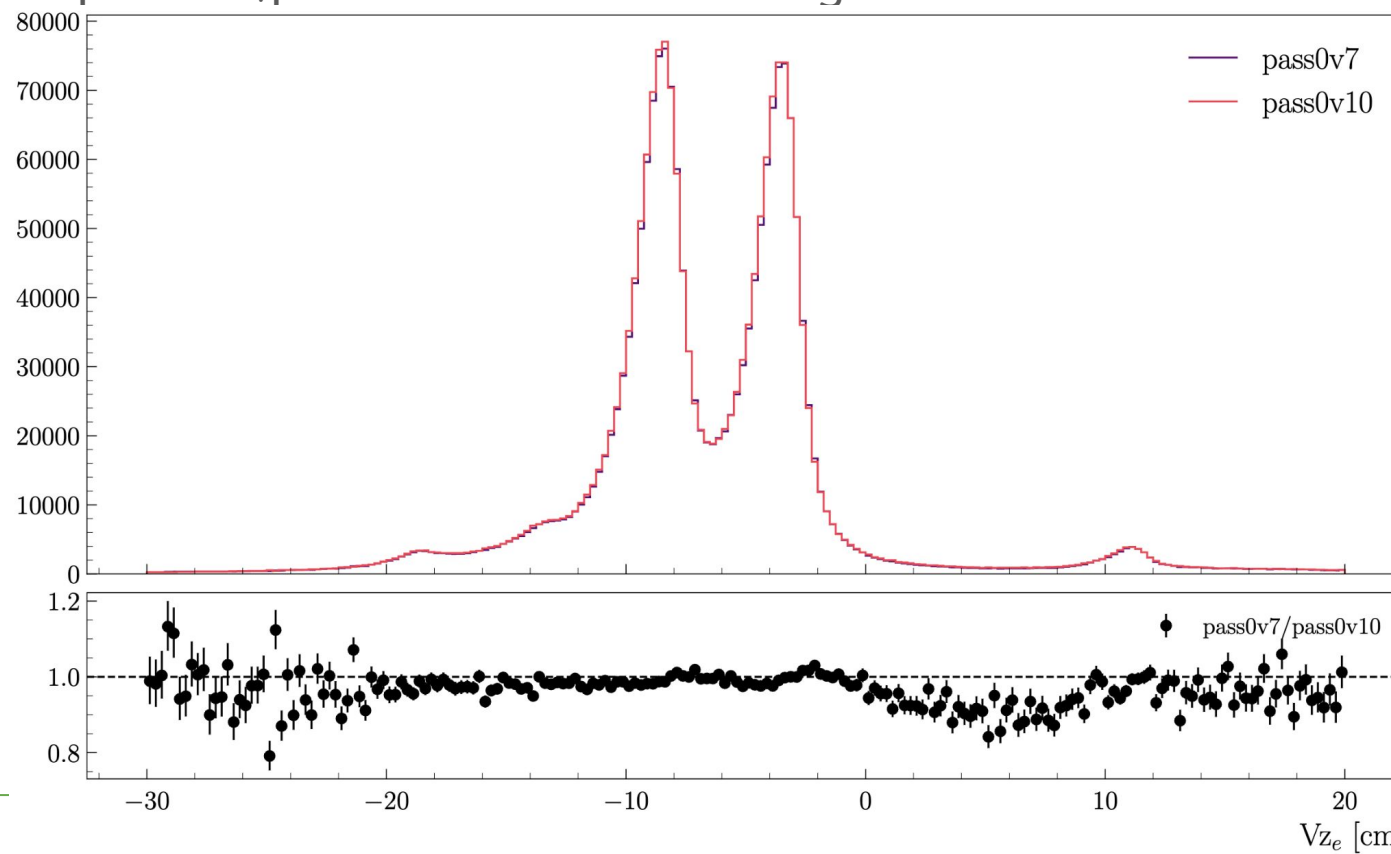


- Some strange runs like 19038 which have a negative slope that are close to a change of the HWP: 18886, 18831, 19038
- CCDB and comment of the logbook say the same information
- But a negative BSA could indicate a issue with the HWP, because it should be positive.
- But we don't have enough statistic to be sure:
 - To solve that, we can cook each these runs + 1 before and 1 after. Issue with the farm so cooking is a bad idea.
 - We can also keep thing like this and check result after full cooking. If we still see some issue:
 - We can change the HWP in the ccdb and recook these runs (RG-C have done that)
 - We can do nothing and change the helicity value in our analysis code (RG-A have done that)
 - We can write a little program that flig the helicity and write a new hipo file with the corrected helicity
 - Chris is ok with any of these solution.

Comparison pass0v7/pass0v10 CxC outbending



Comparison pass0v7/pass0v10 CxC outbending



To select a good electron, we use `pid=11`, `status<0` and `-5 < chi2 < 5`

Cut	Limits
Charge	-1
Number of Photoelectrons	$N_{ph} > 2$
Min. PCAL Energy	$PCAL_{dep} > 60 \text{ MeV}$
Sampling Fraction vs E_{dep}	$\pm 5\sigma$

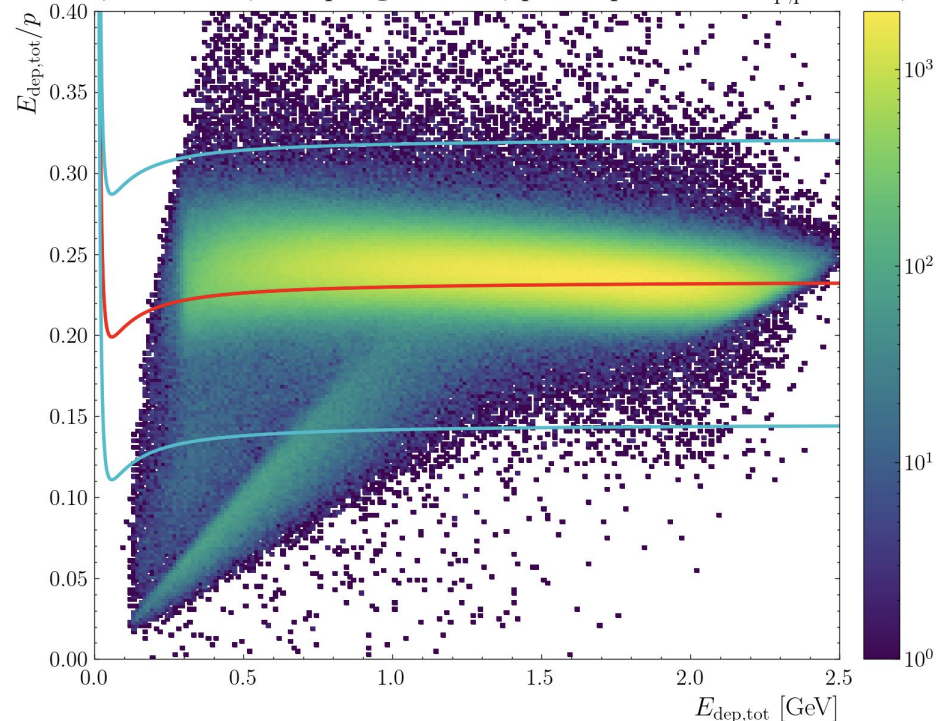
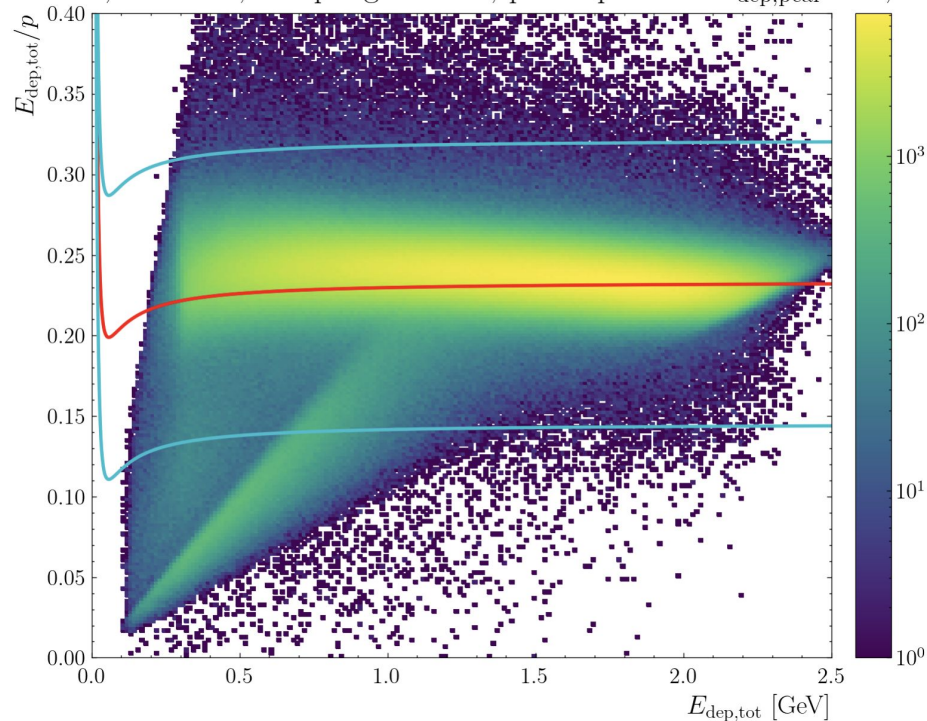
This 5σ cut use a parametrization from CCDB, I asked Raffaella about it and we don't need to extract our own parametrization we can use the one that every other RG use (expect for the runs before 12700)

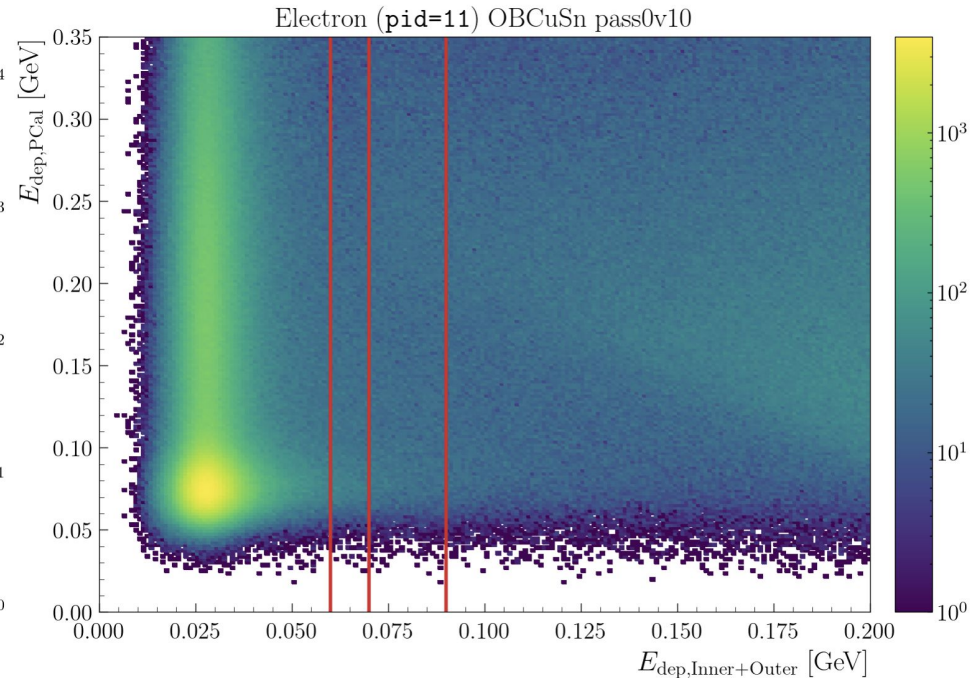
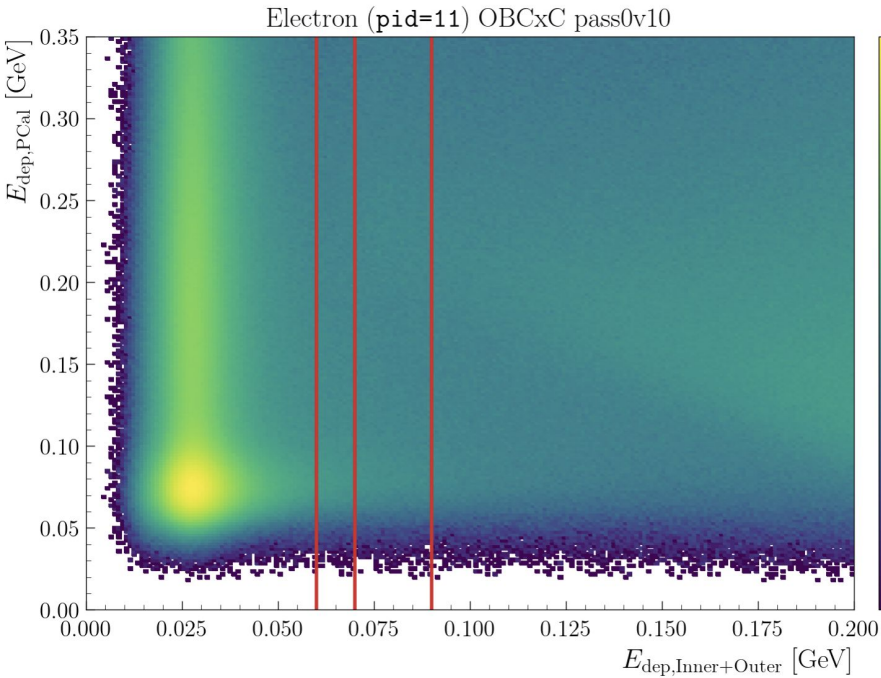
In the common analysis note for RG-A and RG-M, they improve these cuts:

- PCAL Minimum Energy Deposition Cut
- Sampling Fraction vs momentum Cut
- inner calorimeter and the PCAL SF Cut

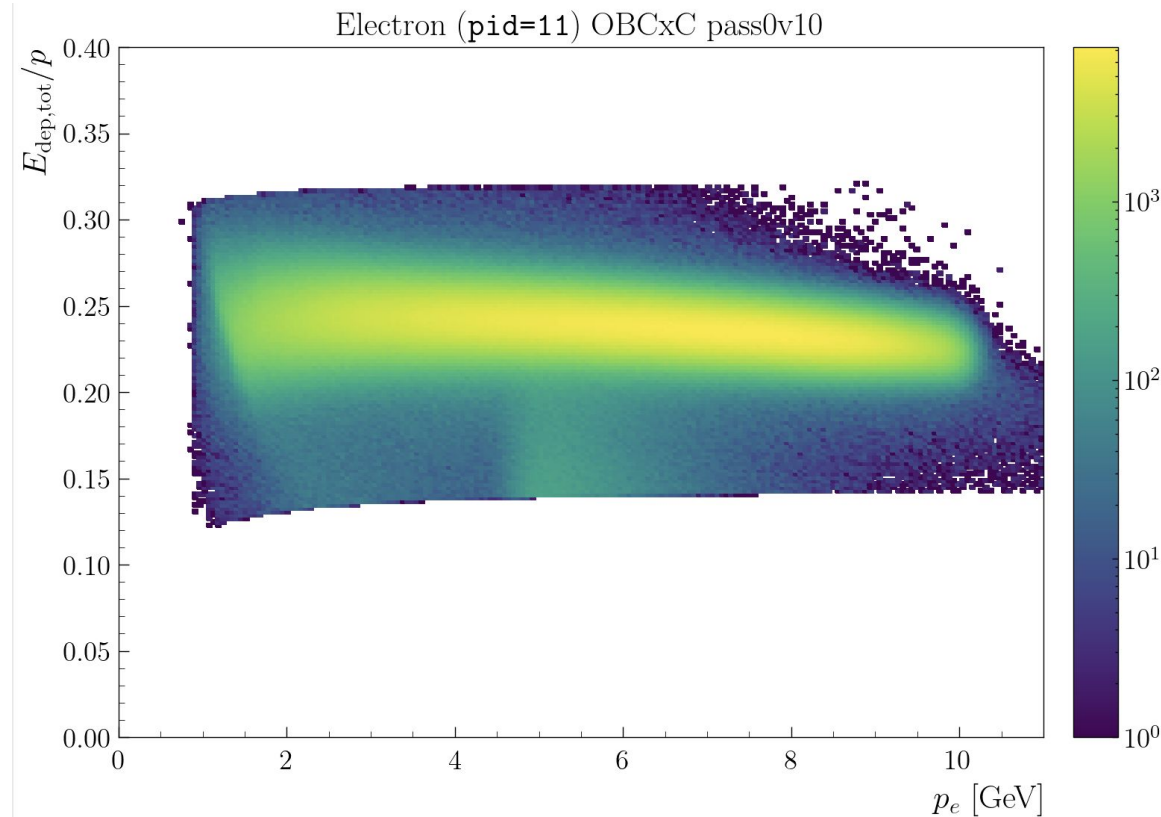
RG-D Analysis: Sampling fraction vs Edep,PCal

pass0v10, OB CxC, Sampling fraction, pass Nphe and $E_{\text{dep,pcal}}$ cuts, S1 pass0v10, OB CuSn, Sampling fraction, pass Nphe and $E_{\text{dep,pcal}}$ cuts, S1





Sampling Fraction vs momentum Cut

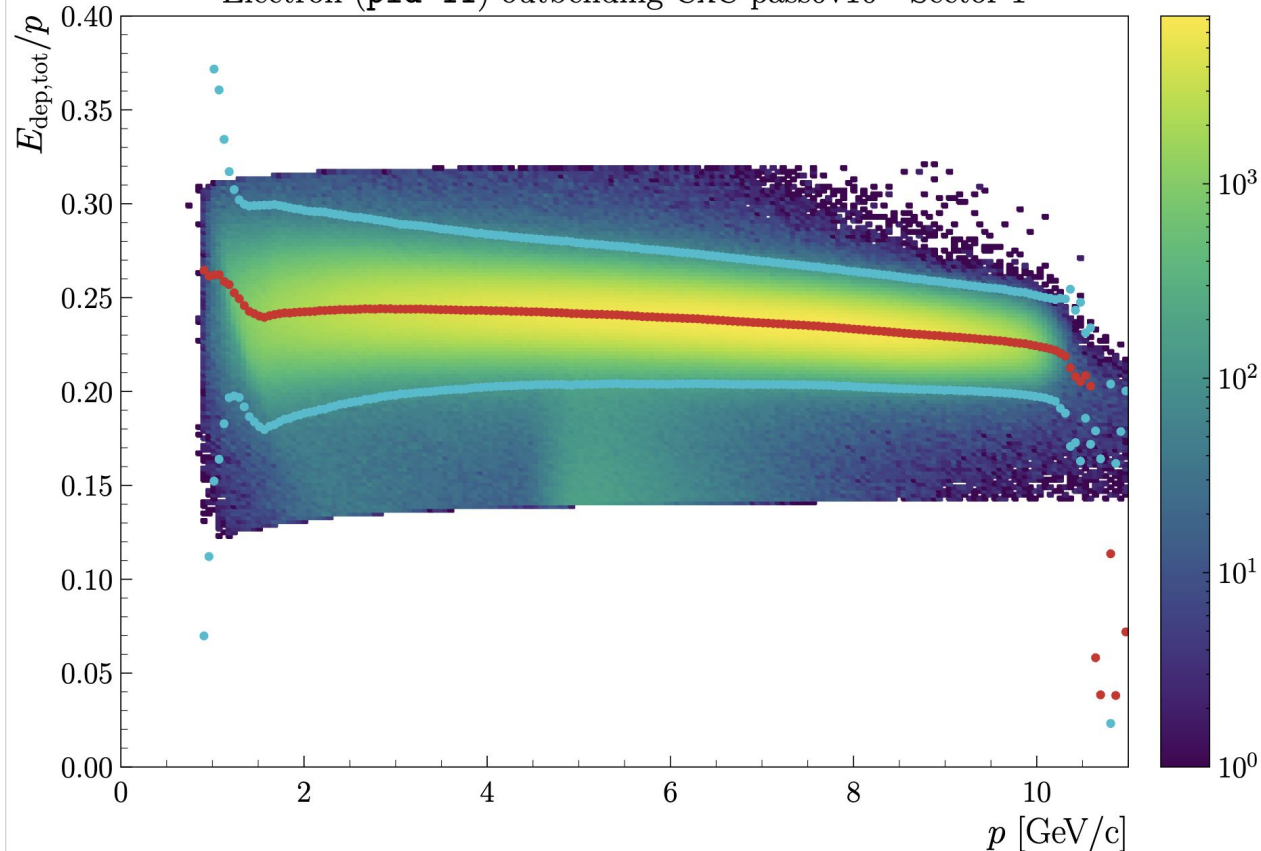


Make 50 slice in momentum (220MeV).

Fit the 1D distribution with a gaussian, extract mean and sigma.

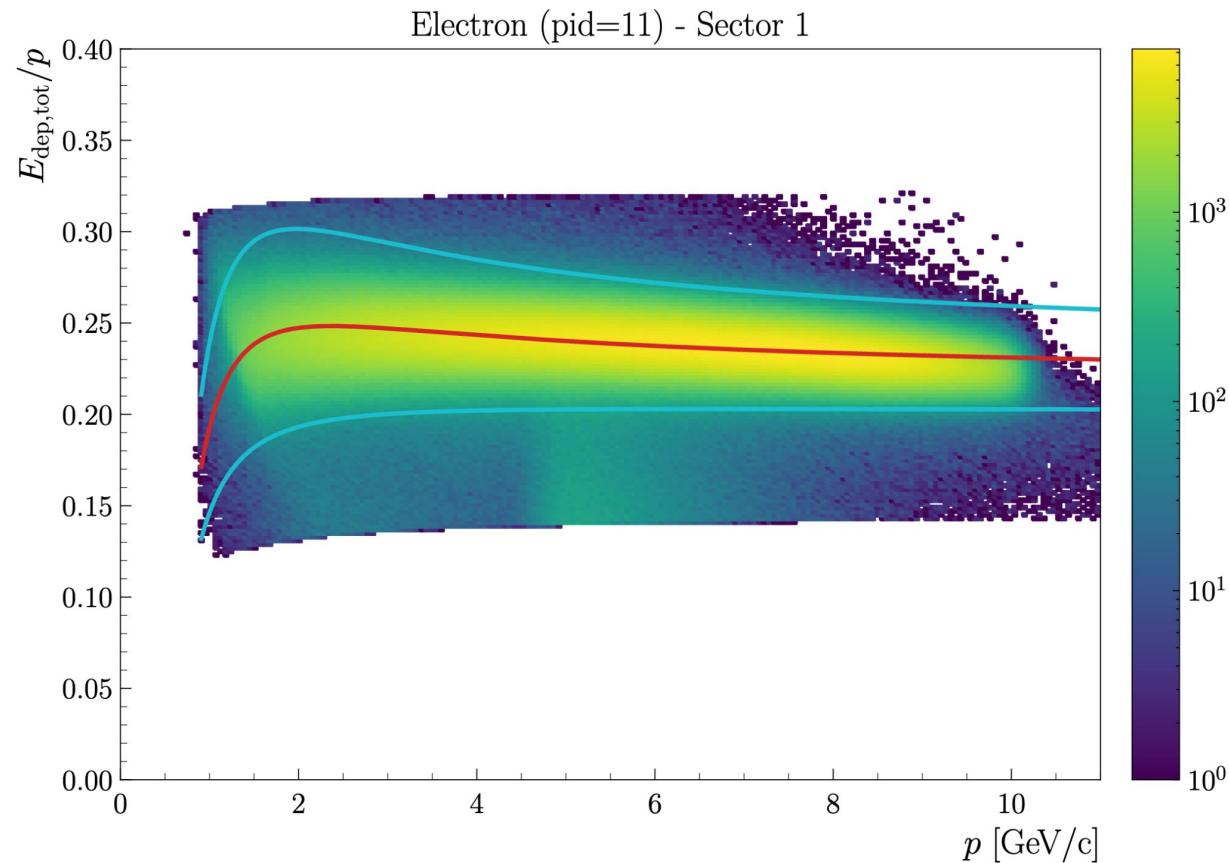
Sampling Fraction vs momentum Cut

Electron ($\text{pid}=11$) outbending CxC pass0v10 - Sector 1



Red is mean, blue is ± 3 sigma.

Sampling Fraction vs momentum Cut



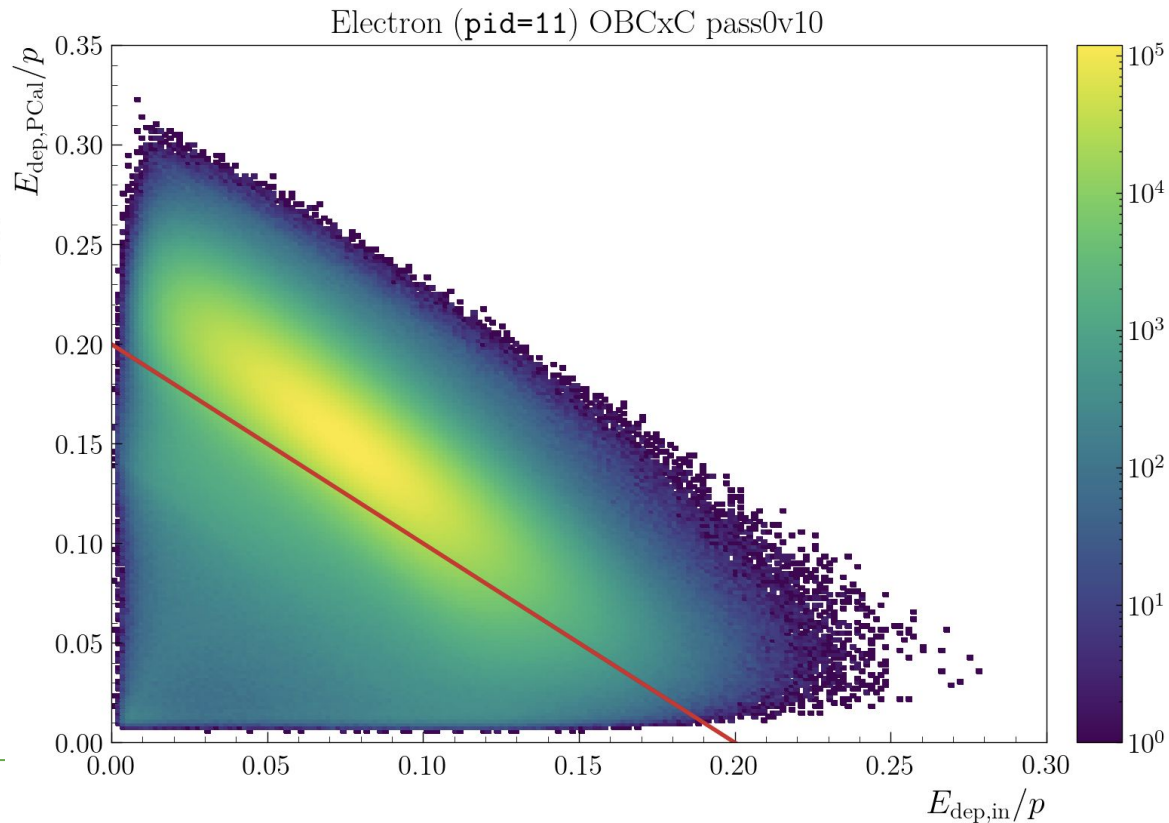
Red is mean, blue is ± 3 sigma.

Fit it with a function:
 $a + b/p + c/p^2$

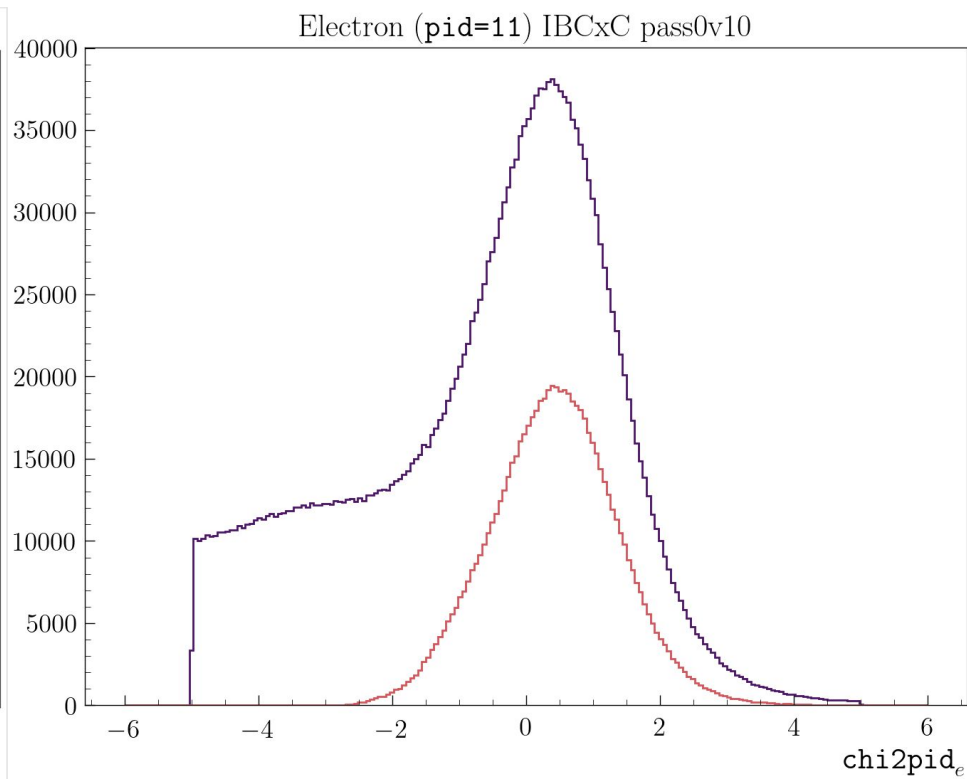
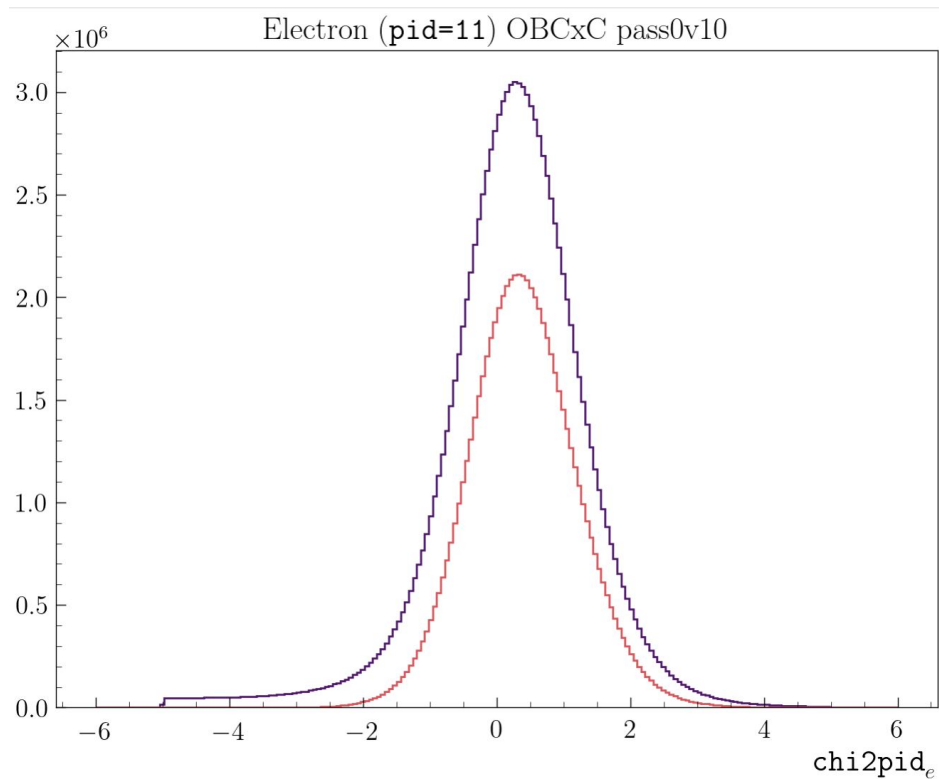
Remove most of the remaining pion contamination above the HTCC.

If $p_e > 4.5$ GeV, apply this cut:

$$\frac{E_{inner}}{p} < 0.2 - \frac{E_{PCAL}}{p}$$

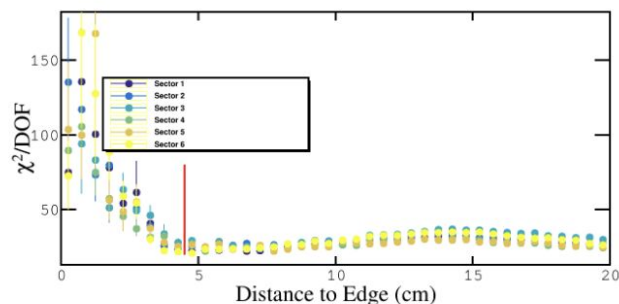
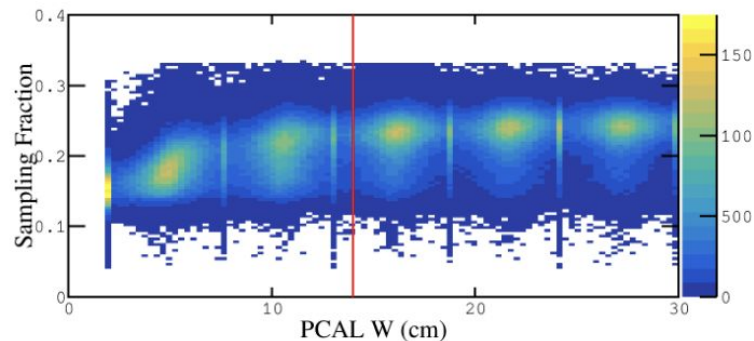
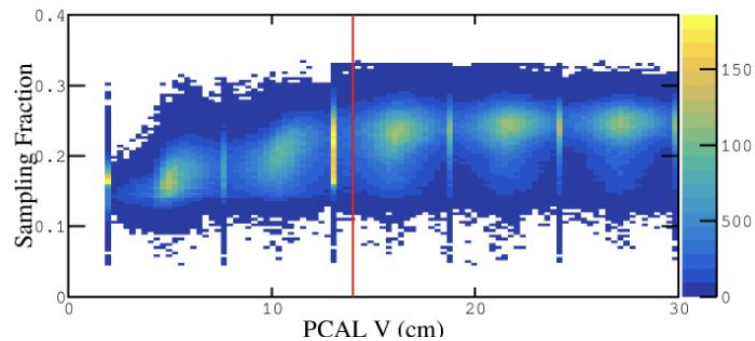


Impact on chi2pid



Remaining cuts

Fiducial cuts: on the calorimeter and the DC (RG-M plots)



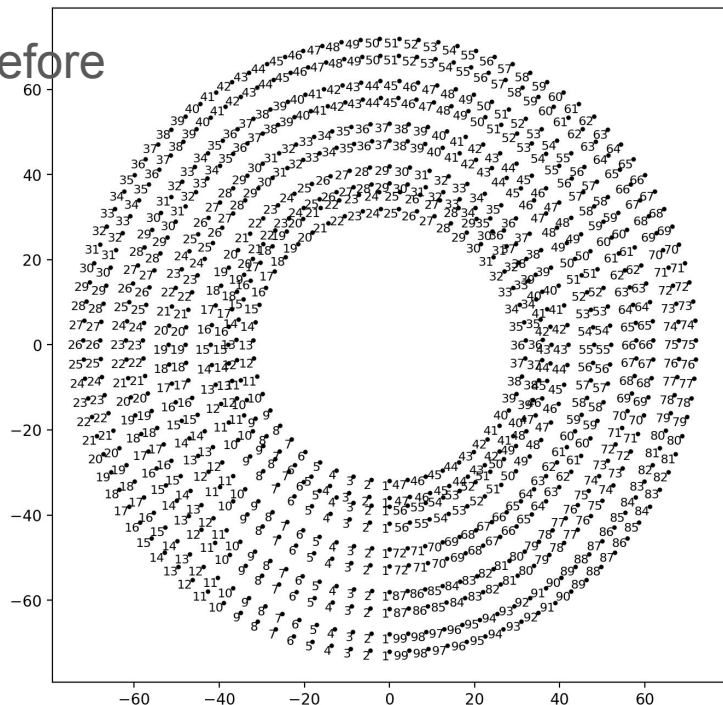
(a) Region 1

I can do that, the v and w are available in the calorimeter bank.
 χ^2/NDF come from the REC::Track and edge
 come from REC::Traj bank

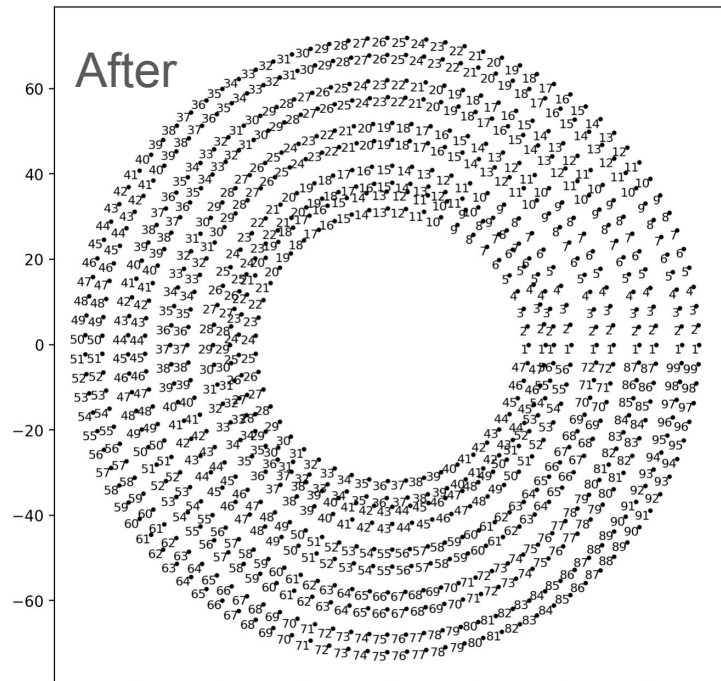
ALERT

Last week, we discover that the geometry in the simulation and the reconstruction is different than the real one.

Before



After



Change geometry for the hits to match reality. (90 shift + counterclock) #544

Last week, we discover that the geometry in the simulation and the reconstruction is different that the real one.

There are still some issues, Raffaella work on that.

Task	expected	done	comment
AI network + validation	04/21		
Vz cuts for pass0v10	-	04/14	
Redo the background merging studies	04/25		
improve pid, fiducials cuts	04/21		
Finish the event mixing method	05/04		
Extract the multiplicity ratio, pT	05/15		
improve code that generated all track candidates (ALERT)	05/15		
Test the AI on real data (ALERT)	04/21		